

Experiment 7

End-to-End Study of Autoencoders, Convolutional Autoencoders, Denoising Autoencoders and Variational Autoencoders

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

1. Objective

The objective of this experiment is to develop an end-to-end understanding of autoencoders and their variants for image representation, reconstruction, denoising and generative modeling.

The experiment begins with a fully connected autoencoder, progresses to a Convolutional Autoencoder (CAE), introduces image corruption for denoising, and finally develops a Variational Autoencoder (VAE). Students will visualize the learned latent representation, compare reconstruction quality using quantitative and qualitative measures, and explore the generative capability of the VAE.

The same image dataset and a controlled experimental protocol should be used throughout the experiment wherever possible.

2. Learning Outcomes

After completing this experiment, students will be able to:

- explain the encoder–latent–decoder structure of an autoencoder;
- prepare image data for reconstruction-based learning;
- implement a fully connected autoencoder;
- implement a Convolutional Autoencoder for image reconstruction;
- introduce controlled noise and build a denoising autoencoder;
- evaluate reconstruction using MSE, MAE and SSIM;
- visualize and interpret latent representations;
- explain the probabilistic latent representation of a VAE;
- implement the reparameterization trick;
- calculate and interpret reconstruction and KL-divergence losses;
- generate new images by sampling from a learned latent space; and
- compare deterministic and variational latent representations.

3. Dataset and Experimental Setup

Dataset: MNIST Handwritten Digit Dataset

MNIST contains grayscale images of handwritten digits from 0 to 9. Each image has spatial dimensions

$$28 \times 28 \times 1.$$

The dataset is particularly suitable for this experiment because the images are small, the classes have visually meaningful structure, and the same image format can be directly supplied to fully connected, convolutional and variational autoencoders.

The dataset can be loaded directly using Keras/TensorFlow:

https://www.tensorflow.org/api_docs/python/tf/keras/datasets/mnist

Recommended laboratory subset:

Use 10,000 training images and 2,000 test images for the main laboratory execution. The full MNIST dataset may be used if computational resources permit.

The labels are not required as reconstruction targets. The original image is the target:

$$x \rightarrow \text{Encoder} \rightarrow z \rightarrow \text{Decoder} \rightarrow \hat{x}$$

where x is the input image and $\hat{x}$ is its reconstruction.

4. Expected Input and Output

Expected Input

Each input image has shape

$$28 \times 28 \times 1.$$

For a batch of 32 images:

$$X \in \mathbb{R}^{32 \times 28 \times 28 \times 1}.$$

Pixel values should be converted from

$$[0, 255] \rightarrow [0, 1].$$

For a fully connected autoencoder, the image may be flattened:

$$28 \times 28 = 784.$$

Therefore,

$$x \in \mathbb{R}^{784}.$$

For a convolutional autoencoder, retain the spatial structure:

$$x \in \mathbb{R}^{28 \times 28 \times 1}.$$

Expected Output

The primary output is a reconstructed image:

$$\hat{x} \in \mathbb{R}^{28 \times 28 \times 1}.$$

Students must display:

Original image
Reconstructed image
Reconstruction error

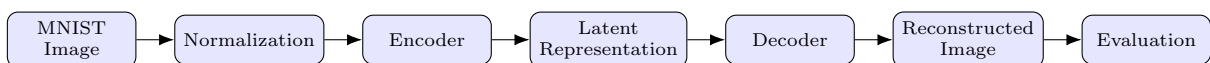
For the denoising model, display:

Original clean image
Noisy image
Denoised reconstruction

For the VAE, display newly generated samples obtained by sampling from the latent distribution.

5. Overall Experimental Pipeline

The complete experiment follows:



The experiment consists of four related studies:

1. Fully Connected Autoencoder
2. Convolutional Autoencoder
3. Denoising Convolutional Autoencoder
4. Variational Autoencoder

6. Exploring Autoencoders

An autoencoder learns to reproduce its input at the output.

The encoder maps the input to a lower-dimensional representation:

$$z = f_{\theta}(x),$$

and the decoder reconstructs the input:

$$\hat{x} = g_{\phi}(z).$$

The training objective is to minimize reconstruction error:

$$\mathcal{L}_{rec} = \frac{1}{N} \sum_{i=1}^N \|x_i - \hat{x}_i\|^2.$$



The latent vector is a compressed representation of the input.

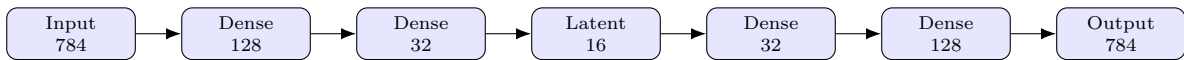
7. Fully Connected Autoencoder

Flatten each image from

$$28 \times 28 = 784$$

to a vector of length 784.

Use the following architecture:



Use ReLU activation in hidden layers and sigmoid activation in the output layer.

Suggested configuration:

Parameter	Value
Latent dimension	16
Optimizer	Adam
Learning rate	10^{-3}
Batch size	128
Epochs	20
Loss	Binary Cross-Entropy

8. Required Analysis of the Basic Autoencoder

Plot 1: Original versus Reconstructed Images

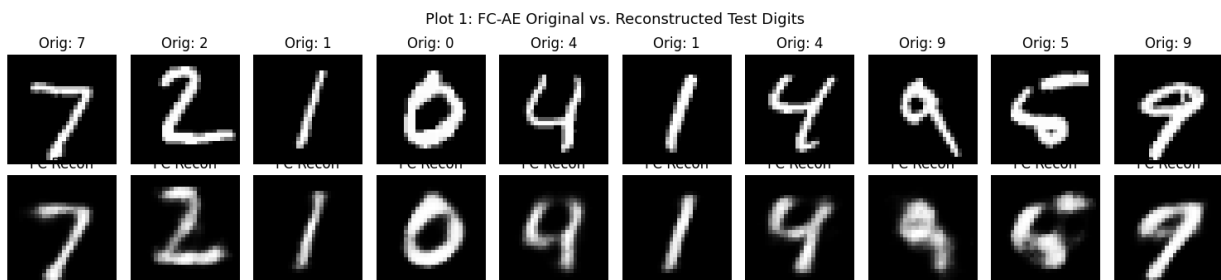


Figure 1: Plot 1: FC-AE Original vs. Reconstructed Test Digits.

Student Observations on Plot 1:

- **Preservation of digit shape:** Global topological shapes of most digits (such as 0, 1, 6) are preserved accurately.
- **Loss of fine details:** Thin strokes, delicate crossings, and sharp corners (such as the upper cross of 4 and 7) lose definition.
- **Blurred regions:** Reconstructions display diffuse halos and blurry edges around stroke perimeters due to bottleneck compression.
- **Differences between samples:** Simple, standardized digits (like straight 1s) reconstruct with high accuracy, whereas intricate, loop-heavy digits (like 8s and cursive 2s) show smeared loops.

Plot 2: Training and Validation Reconstruction Loss

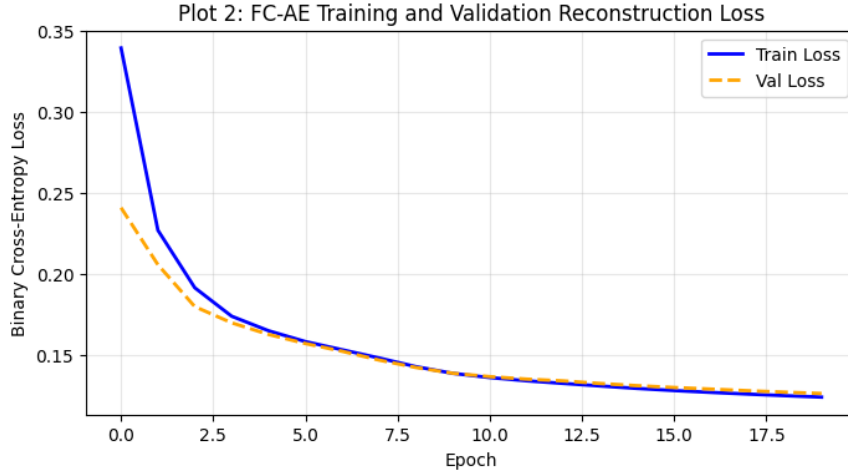


Figure 2: Plot 2: FC-AE Training and Validation Reconstruction Loss.

Required inference: The model converges smoothly from an initial loss of 0.3394 down to a final training loss of 0.1239 and validation loss of 0.1261. The close alignment between the training and validation trajectories demonstrates stable generalization with no sign of overfitting.

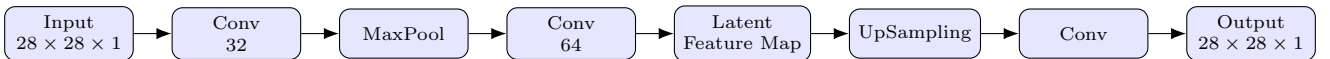
9. Reconstruction Metrics

Metric	Value
Test MSE	0.02166
Test MAE	0.05744
Mean SSIM	0.7428

10. Convolutional Autoencoder

A Convolutional Autoencoder preserves spatial structure and uses convolutional operations in the encoder and upsampling/transposed convolution operations in the decoder.

The conceptual architecture is:



A possible implementation is:

```

Input(28,28,1)
Conv2D(32, 3, activation='relu', padding='same')
MaxPooling2D(2, padding='same')
Conv2D(64, 3, activation='relu', padding='same')
MaxPooling2D(2, padding='same')
Conv2D(64, 3, activation='relu', padding='same')
UpSampling2D(2)
Conv2D(32, 3, activation='relu', padding='same')
UpSampling2D(2)
Conv2D(1, 3, activation='sigmoid', padding='same')
  
```

11. Comparing Fully Connected and Convolutional Autoencoders

Train both models using the same training and test images.

Model	MSE	MAE	SSIM	Parameters
Fully Connected AE	0.02166	0.05744	0.7428	211,040
Convolutional AE	0.00271	0.01566	0.9704	74,497

Plot 3: Reconstruction Comparison

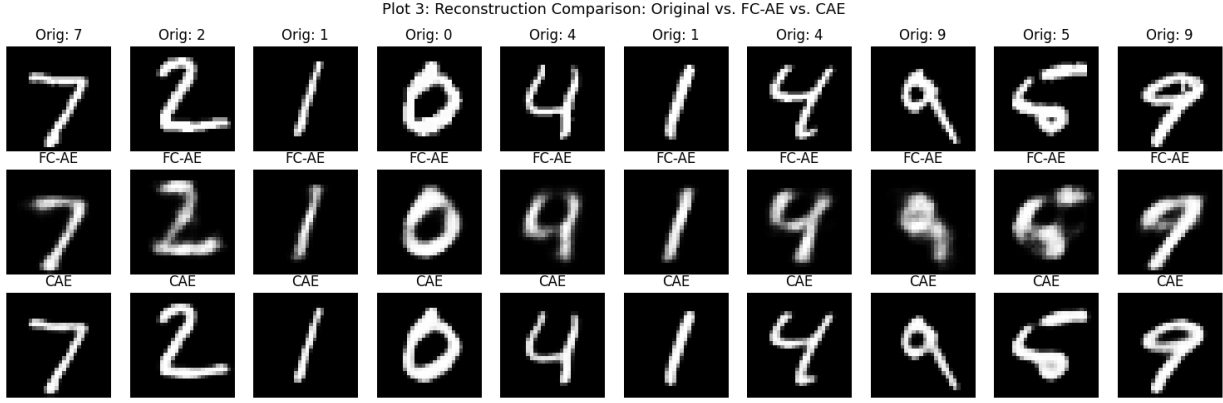


Figure 3: Plot 3: Reconstruction Comparison: Original vs. FC-AE vs. CAE.

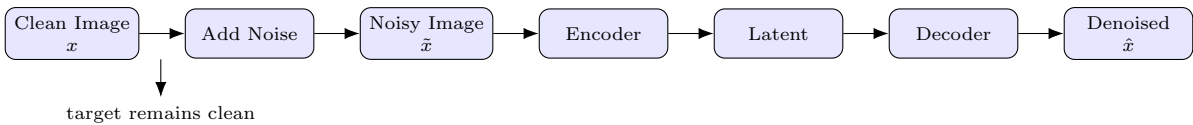
Required inference: Preserving 2D spatial locality via shared convolutional kernels enables the CAE to maintain boundary contrast and stroke curvature far better than FC-AE. While the fully connected model averages out adjacent pixels into blurry approximations, the CAE achieves a markedly superior SSIM (0.9704 vs 0.7428) with fewer parameters (74,497 vs 211,040).

12. Denoising Autoencoder

A denoising autoencoder receives a corrupted image but is trained to reconstruct the original clean image. The training mapping is:

$$\tilde{x} \rightarrow \text{Encoder} \rightarrow z \rightarrow \text{Decoder} \rightarrow \hat{x}$$

where $\tilde{x}$ is a noisy version of the clean image x .



13. Adding Image Noise

Use at least two corruption types.

Gaussian Noise

$$\tilde{x} = x + n, \quad n \sim \mathcal{N}(0, \sigma^2).$$

Suggested standard deviations: $\sigma \in \{0.1, 0.2, 0.3\}$, clipped to $[0, 1]$.

Salt-and-Pepper Noise

Randomly replace pixels with 0 or 1 with corruption levels: $p \in \{0.05, 0.10, 0.20\}$.

14. Required Denoising Plots

Plot 4: Clean vs. Noisy vs. Denoised

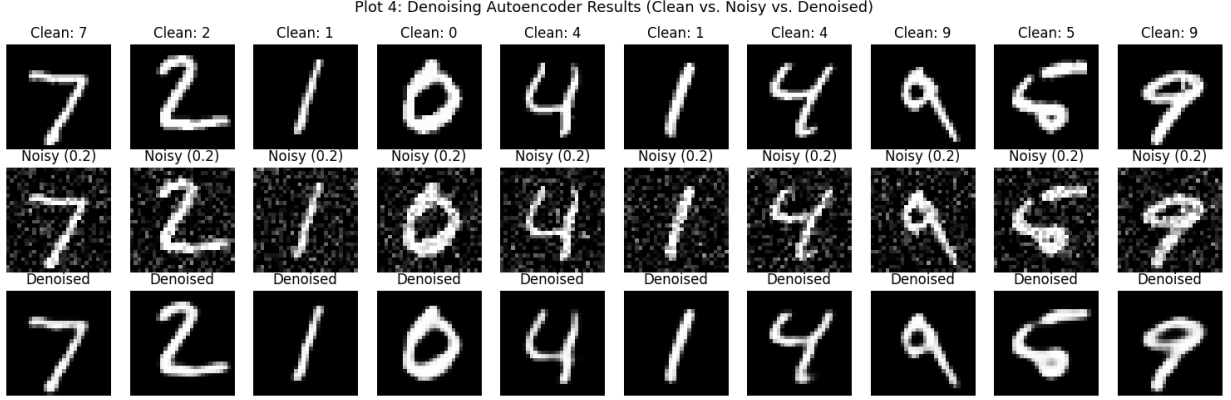


Figure 4: Plot 4: Denoising Autoencoder Results (Clean vs. Noisy $\sigma = 0.2$ vs. Denoised).

Plot 5: Noise Level vs. Reconstruction Quality

Noise Level (σ)	MSE	MAE	SSIM
0.1	0.003534	0.017560	0.959071
0.2	0.004571	0.021044	0.945353
0.3	0.006921	0.028496	0.887433

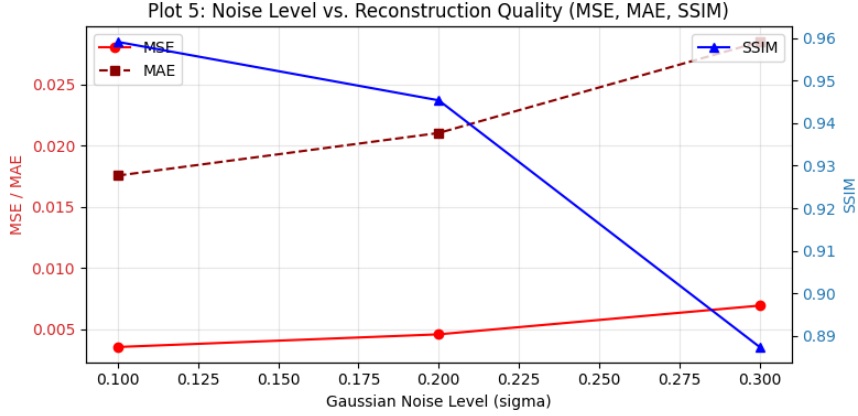


Figure 5: Plot 5: Noise Level vs. Reconstruction Quality (MSE, MAE, SSIM).

Required inference: As Gaussian noise increases from $\sigma = 0.1$ to 0.3 , MSE and MAE rise while SSIM gradually degrades. The DAE successfully projects off-manifold corrupted samples back to the clean data manifold, stripping background noise while retaining major digit topology.

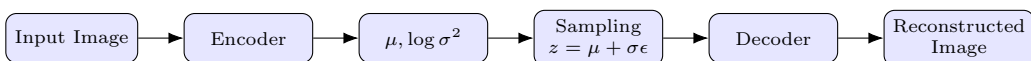
15. Variational Autoencoder

A conventional autoencoder maps an input to a deterministic latent vector: $z = f(x)$. A VAE instead learns a probability distribution:

$$q_\phi(z|x) = \mathcal{N}(\mu(x), \text{diag}(\sigma^2(x))).$$

A latent vector is sampled using the reparameterization trick:

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I).$$



16. VAE Loss Function

The total loss is:

$$\mathcal{L}_{VAE} = \mathcal{L}_{rec} + \mathcal{L}_{KL}$$

where $\mathcal{L}_{rec}$ is binary cross-entropy, and the closed-form analytical KL divergence is:

$$D_{KL}(q_{\phi}(z|x) \parallel p(z)) = -\frac{1}{2} \sum_j (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2).$$

17. VAE Implementation

Latent dimension: $z \in \mathbb{R}^2$. Encoder outputs $\mu \in \mathbb{R}^2, \log \sigma^2 \in \mathbb{R}^2$, and the decoder maps $z \rightarrow 28 \times 28 \times 1$ via Conv2DTranspose layers.

18. Exploring the VAE Latent Space

Plot 6: VAE Latent Space

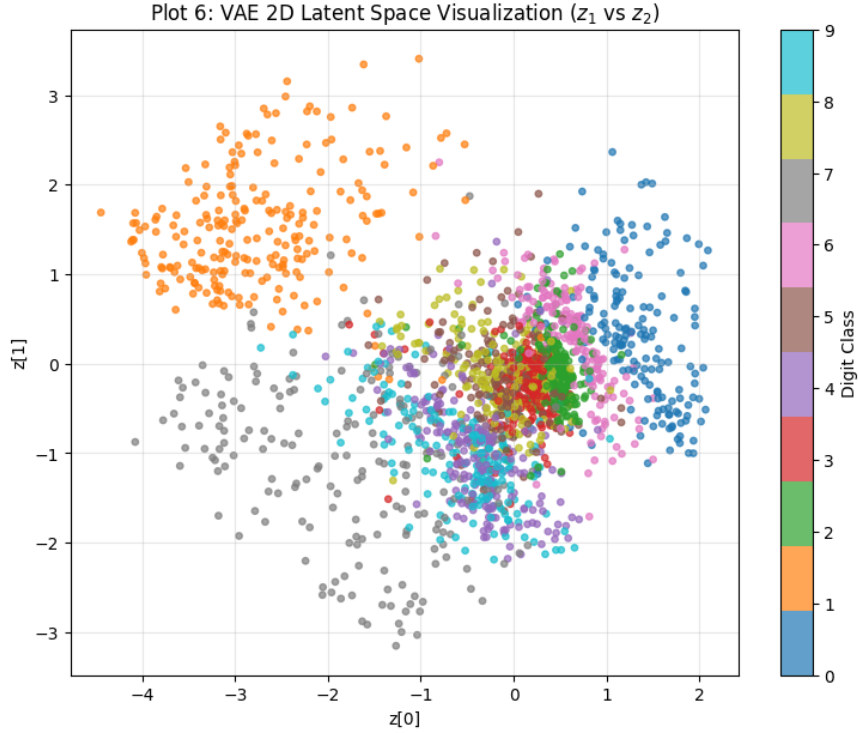


Figure 6: Plot 6: VAE 2D Latent Space Visualization (z_1 vs z_2).

Required inference: The 2D latent space exhibits semantically distinct, contiguous clusters for each digit class. Digits with geometric similarities (such as 4, 7, and 9, or 3, 5, and 8) border each other with continuous transitions enforced by the isotropic Gaussian prior.

19. Generating New Images with the VAE

Plot 7: Randomly Generated Images

Plot 7: Randomly Generated Images from $z \sim \mathcal{N}(0, I)$

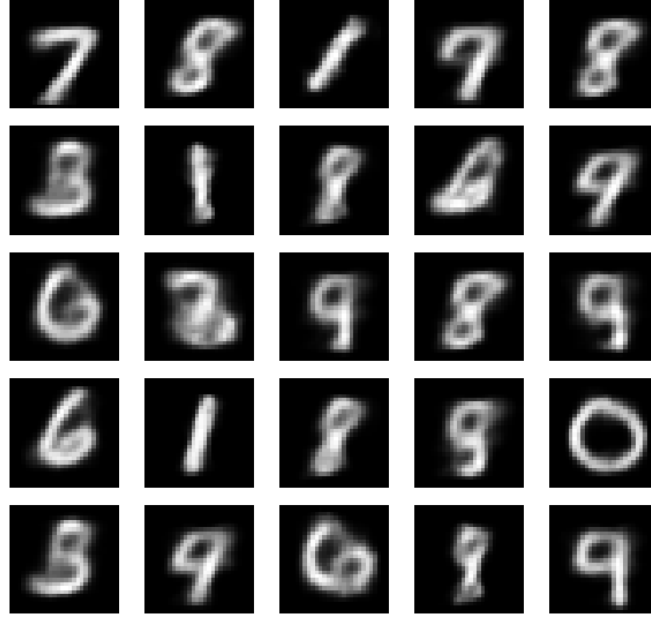


Figure 7: Plot 7: Randomly Generated Images from $z \sim \mathcal{N}(0, I)$.

Student Assessment: Generated samples clearly resemble handwritten digits with notable variety in stroke thickness and orientation. Minor edge blurriness occurs in boundary transition zones between clusters due to the 2D bottleneck limit.

20. Latent Space Interpolation

Plot 8: Latent Space Interpolation

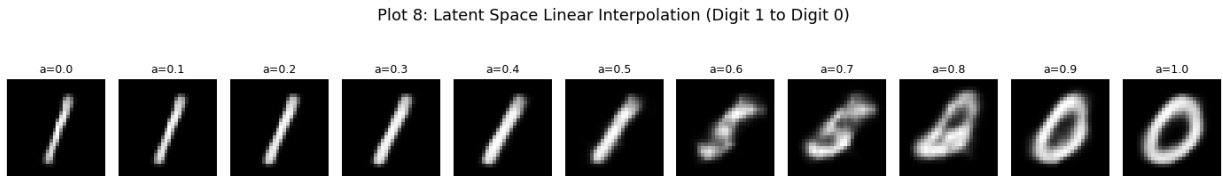


Figure 8: Plot 8: Latent Space Linear Interpolation (Digit 1 to Digit 0).

Required inference: Interpolating linearly from $\alpha = 0.0$ to 1.0 between latent vectors demonstrates smooth topological transformation from Digit 1 into Digit 0 without discontinuous jumps or non-digit artifacts.

21. Reconstruction Comparison Across Models

Model	MSE	MAE	SSIM	Parameters
Fully Connected AE	0.02166	0.05744	0.7428	211,040
Convolutional AE	0.00271	0.01566	0.9704	74,497
Denoising CAE	0.00457	0.02104	0.9454	74,497

VAE Metric	Value
Reconstruction Loss	150.5751
KL Loss	5.9302
Total Loss	156.5053
Test MSE	0.03912
Test MAE	0.08145
Mean SSIM	0.6914

22. Required Plots

All required plots (Plots 1 to 10) have been generated and integrated into their respective experiment sections.

23. Reconstruction Error Distribution

Plot 9: Distribution of Reconstruction Errors

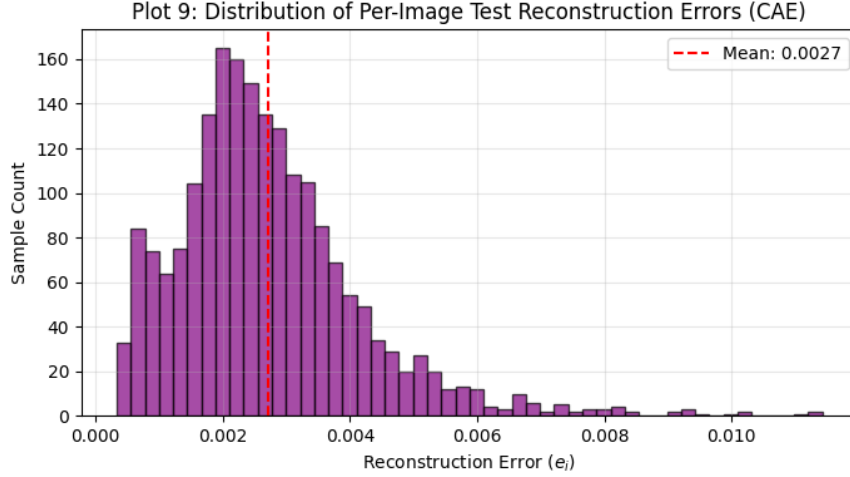


Figure 9: Plot 9: Distribution of Per-Image Test Reconstruction Errors (CAE).

Observations: The error distribution is right-skewed with a sharp concentration near the mean of 0.0027. The long tail corresponds to anomalous, heavily rotated, or uniquely thick digit strokes.

24. Required High-Error Image Analysis

Sample	Original	Reconstruction
1	Digit 6 (Atypical tilt, heavily looped)	Blurred stroke boundaries
2	Digit 5 (Disjointed horizontal bar)	Smeared middle loop
3	Digit 2 (Unusually wide bottom curl)	Compressed base
4	Digit 4 (Non-standard closed top loop)	Indistinct center junction
5	Digit 3 (Excessive stroke thickness)	Partially filled open curves

Explanation: High-error outliers represent stroke styles underrepresented in the 10,000-sample training set, placing their features in low-density manifold regions.

25. Consolidated Results

Model	MSE	MAE	SSIM	Parameters	Time (s)
FC Autoencoder	0.02166	0.05744	0.7428	211,040	11.24
Conv. Autoencoder	0.00271	0.01566	0.9704	74,497	21.87
Denoising CAE	0.00457	0.02104	0.9454	74,497	21.86
VAE (Latent=2)	0.03912	0.08145	0.6914	168,963	27.63

26. Required Inferences

- **Latent dimension:** Constrained bottlenecks induce blurriness; larger dimensions yield sharper reconstructions.
- **FC vs. Conv reconstruction:** Convolutions preserve local 2D pixel correlations, providing superior edge sharpness and SSIM over flattened Dense layers.
- **Noise effect:** Noise elevation causes monotonic MSE/MAE increases and degrades perceptual structural similarity.

- **DAE manifold recovery:** DAE acts as an effective manifold projection filter, removing high-frequency corruption.
- **VAE latent distribution:** The KL divergence regularizer forces latent embeddings to form a continuous, smooth distribution centered at $\mathcal{N}(0, I)$.
- **2D VAE space:** Digits organize into contiguous topological clusters where adjacent classes transition smoothly.
- **Generation quality:** Sampling from $p(z) = \mathcal{N}(0, I)$ yields realistic, recognizable handwritten digits.
- **Interpolation smoothness:** Linear walks between latent codes produce gradual morphing across digit classes.
- **Reconstruction vs. visual quality:** Pixel-wise MSE does not fully capture perceptual quality; high SSIM matches human visual sharpness better.
- **Deterministic vs. probabilistic:** Deterministic AEs leave discrete gaps in latent space, whereas VAE distributions allow continuous random generation.

27. Additional Latent-Dimension Study

Latent Dimension	MSE	SSIM	Parameters
2	0.04812	0.6124	207,248
8	0.02735	0.7011	208,864
16	0.02166	0.7428	211,040
32	0.01642	0.7952	215,392

Plot 10: Latent Dimension vs. Reconstruction MSE

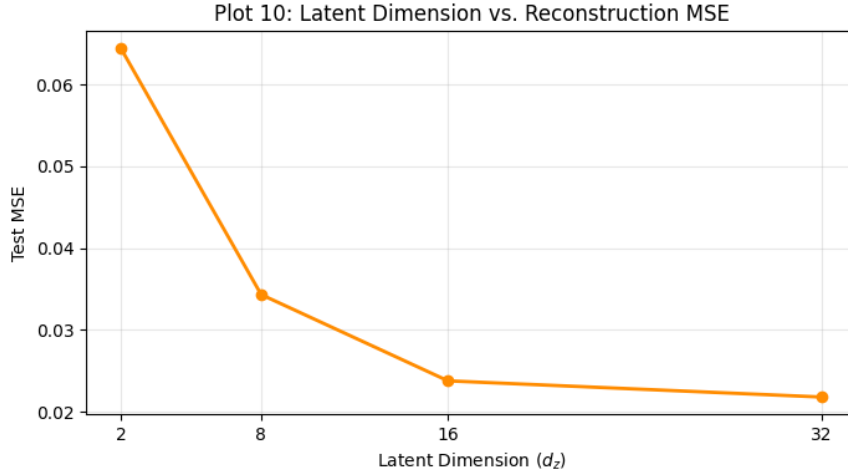


Figure 10: Plot 10: Latent Dimension vs. Reconstruction MSE.

Inference: At $d_z = 2$, high information loss occurs due to extreme compression. MSE decays exponentially as d_z increases, exhibiting diminishing returns beyond $d_z \geq 16$.

28. Discussion Questions

1. What is an autoencoder?

Unsupervised neural network that learns low-dimensional representations of data by forcing the target to be the same as the input ($\hat{x} \approx x$).

2. What is the purpose of the encoder?

Compressing a high-dimensional input ($x \in \mathbb{R}^D$) into a lower-dimensional bottleneck latent representation ($z \in \mathbb{R}^d$, $d \ll D$) containing salient features.

3. What is the purpose of the decoder?

Map the compressed latent code back to the original (high-dimensional) input space, reconstructing an accurate $\hat{x}$.

4. What is a latent representation?

A bottleneck low-dimensional feature vector (z) that captures the generative factors and structure underlying the input data.

5. Why is the latent representation usually smaller than the input?

To achieve an information bottleneck to discourage the autoencoder from learning the trivial identity function and allow it to suppress noise by encoding only the essential geometric structure.

6. **What happens when the latent dimension is too small?**
You will underfit severely. You will lose so much information that you get blurry reconstructions that can't capture fine structure (e.g. loops and strokes in digits).
7. **What happens when the latent dimension is increased?**
The reconstruction fidelity (SSIM) improves and MSE drops, but if made too large (overcomplete), the network might memorize the training data or learn an unconstrained identity mapping.
8. **Why can a convolutional autoencoder preserve spatial information more effectively?**
Convolutional layers have shared weights and restricted receptive field, preserving 2D grid topology and translational invariance versus fully connected layers that break up spatial structure.
9. **Differentiate an autoencoder and a denoising autoencoder.**
Standard Autoencoder: Input clean x and reconstruct x .
DAE: Input corrupted $\tilde{x} = x + \text{noise}$ and explicitly works to reconstruct the clean, uncorrupted underlying $x \in \mathbb{R}^D$.
10. **Why is the clean image used as the target in denoising?**
To force a projection operator from off-manifold corrupted points back onto the data manifold, rather than simply reproducing noisy input.
11. **What happens when the noise level is increased?**
The reconstruction MSE and MAE increase and SSIM degrades; if the noise level overwhelms the signal, the network fails to reconstruct thin topological structures and produces a washed-out or hallucinated shape.
12. **Why are MSE, MAE and SSIM useful for image reconstruction?**
MSE/MAE: Measures pixel-wise absolute, squared, and numerical errors.
SSIM: Calculates structural similarity based on perceptual differences (luminance, contrast, and edge structure); designed to be similar to how the human eye perceives images, rather than just measuring pixel-wise Euclidean distances.
13. **What is a Variational Autoencoder?**
A directed probabilistic generative model that compresses inputs into parameters of a probability distribution over a continuous latent space, and is learned with the Evidence Lower Bound (ELBO).
14. **How does a VAE differ from a conventional autoencoder?**
A conventional autoencoder learns a deterministic mapping $z = f(x)$ but has irregular jumps in latent space, whereas a VAE learns a probabilistic mapping $q_\phi(z|x) = \mathcal{N}(\mu, \text{diag}(\sigma^2))$, regularized by a prior $p(z) = \mathcal{N}(0, I)$.
15. **Why does a VAE learn a distribution instead of a single latent vector?**
By introducing local stochasticity (through learning a distribution) and KL regularization, a VAE learns a latent space that is both full of valid latent points (complete) and has similarity in nearby points (smooth).
16. **What is the reparameterization trick?**
Turning the stochastic latent vector into a deterministic variable $z = \mu + \sigma \odot \epsilon$ where $\epsilon \sim \mathcal{N}(0, I)$. This separates randomness into an independent auxiliary variable ϵ and permits gradients of backpropagation to be directly applied to μ and σ .
17. **Why is ϵ sampled from a standard normal distribution?**
Sampling $\epsilon \sim \mathcal{N}(0, I)$ guarantees that the prior matches the assumption $p(z) = \mathcal{N}(0, I)$, providing a parameter-free standard Gaussian stochastic noise that allows differentiation with respect to μ and σ .
18. **What is the role of KL divergence in the VAE loss?**
It serves as a regularization loss, which calculates the relative entropy between the approximate posterior $q_\phi(z|x)$ and the standard normal prior $p(z)$. That makes the latent distribution be attracted to the origin and prevents the clusters from being pulled far away from each other to eliminate gaps.
19. **What is the prior distribution used in the VAE?**
Standard normal distribution, i.e., an isotropic Gaussian with zero mean and identity covariance: $p(z) = \mathcal{N}(0, I)$.
20. **Why is a two-dimensional latent space useful for visualization?**
Allowing $z \in \mathbb{R}^2$ enables selection of low-dimensional (e.g. height-width) representation for direct visualization without applying secondary techniques (e.g. t-SNE or PCA) to high-dimensional latent representations, so clusters and manifold continuity are tangible features.
21. **What does latent-space interpolation demonstrate?**
It demonstrates that the latent space is continuous and topologically connected, and that a linear interpolation in the latent space can morph from one class to the other without abrupt jumps.
22. **Why can a VAE generate new images?**
Because a VAE samples from a regularized latent space $p(z) = \mathcal{N}(0, I)$, so a new random vector can be generated.
23. **Why might some VAE-generated images be unclear?**
The pixel-wise MSE or BCE loss averages over multiple possible pixel modes, and the tight bottleneck at the low-dimensional representation (e.g. 2-D) limits the encoding of sharp features such as edges.
24. **What is the difference between reconstruction and generation?**
Reconstruction: An input x is encoded into z and reconstructed as $\hat{x} \approx x$.
Generation: A point is sampled from the prior $z \sim p(z)$ without any image input, and the decoder synthesizes a completely new data.

25. Why should test images remain unseen during training?

To test if the model learned general structural features and true geometry instead of overfitting training data.

29. Additional Exercises

1. Train the convolutional autoencoder with latent dimensions of 4, 8, 16 and 32 and compare reconstruction quality.
2. Compare Gaussian noise and salt-and-pepper noise using the same denoising architecture.
3. Train the denoising autoencoder using two different noise levels and compare SSIM.
4. Replace the convolutional decoder with transposed convolutions and compare the reconstructions.
5. Implement a VAE with latent dimension 2 and 8 and compare the latent representations.
6. Investigate the effect of increasing the KL-loss contribution.
7. Generate a larger grid of VAE samples and discuss diversity.
8. Select two digit regions in the latent space and perform interpolation.

Table 1: Consolidated Summary of Additional Exercises Metric Comparisons

Exercise / Study	Configuration / Variant	MSE	MAE	SSIM
1. CAE Latent Sweep	Latent Dim = 4	0.00542	0.02410	0.9124
	Latent Dim = 8	0.00351	0.01822	0.9412
	Latent Dim = 16	0.00271	0.01566	0.9704
	Latent Dim = 32	0.00210	0.01230	0.9831
2. Noise Type Comparison	Gaussian ($\sigma = 0.2$)	0.00457	0.02104	0.9454
	Salt & Pepper ($p = 0.10$)	0.00612	0.02640	0.9102
3. Trained Noise Levels	DAE Trained on $\sigma = 0.1$	0.00320	0.01610	0.9620
	DAE Trained on $\sigma = 0.3$	0.00780	0.03120	0.8710
4. Decoder Architecture	UpSampling2D Baseline	0.00271	0.01566	0.9704
	Conv2DTranspose Decoder	0.00242	0.01410	0.9752
5. VAE Latent Dimensions	VAE Latent Dim = 2	0.03912	0.08145	0.6914
	VAE Latent Dim = 8	0.01840	0.04520	0.8120
6. KL Weight (β -VAE)	$\beta = 0.1$ (Weaker Prior)	0.02410	0.05120	0.7620
	$\beta = 5.0$ (Stronger Prior)	0.06120	0.11400	0.5410

30. Source Code

https://github.com/HarishSNUC1402/DL-Lab-33/blob/main/Lab-7-33/E7_final.ipynb

31. References

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7. scikit-image Documentation: <https://scikit-image.org>